

1 Definitions, Classification of PDEs

1.1 Big O, Classification

Big O: $f(t) = \mathcal{O}(g(t))$ as $t \rightarrow t_0$ if $\exists \varepsilon > 0, C > 0$ s.t. $\forall |t - t_0| < \varepsilon, |f(t)| \leq C|g(t)|$
 $\Leftrightarrow \lim_{t \rightarrow t_0} [f(t)/g(t)]$ bdd ps: If not say $t \rightarrow t_0$, assume $t \rightarrow 0$

More on Big O: a. $F(x) = G(x) + \mathcal{O}(H(x))$ if $F(x) - G(x) = \mathcal{O}(H(x))$

b. $F(x) = \mathcal{O}(G(x)) + \mathcal{O}(H(x))$ if $F(x) = \mathcal{O}(\max\{|G(x)|, |H(x)|\})$
 c. $F(x, y) = \mathcal{O}(m^x) + \mathcal{O}(y^n)$ [if Given $G(x) = \mathcal{O}(x^{\frac{m}{n}}) \Rightarrow F(x, G(x)) = \mathcal{O}(x^m)$], 且另一项同理

Classification: For $a u_{xx} + b u_{xy} + c u_{yy} + d u_x + e u_y + f u + g = 0$ ($a, b, c \neq (0, 0, 0)$)

1. **Hyperbolic:** $b^2 - 4ac > 0$ Admits non-trivial plane wave solutions (Propagation)
2. **Parabolic:** $b^2 - 4ac = 0$ and we can get $2cd \neq be$ or $2ae \neq bd$
3. **Elliptic:** $b^2 - 4ac < 0$ No non-trivial plane wave solutions (No Propagation)

Remark: 对变量 x, y 进行缩放后不会改变 PDE 类型

1.2 Normal

Weighted 2-Norm: $\|\mathbf{u}\|_{2, \Delta x} = \sqrt{(\Delta x)^d \|\mathbf{u}\|_2}$ where d s.t. $\mathbf{u} : \mathbb{R}^d \rightarrow \mathbb{R}^n$

- For $d = 1$ and $\Delta x = \frac{D}{N}$, then: $\forall i, u_i = \mathcal{O}((\Delta x)^p) \Rightarrow \|\mathbf{u}\|_{2, \Delta x} = \mathcal{O}((\Delta x)^p)$

- For A symmetric: $\|A^{-1} \mathbf{w}\|_2 \leq \frac{1}{|\lambda_1|} \|\mathbf{w}\|_2$ $0 < |\lambda_1| \leq \dots \leq |\lambda_n|$

Property: If $\|\mathbf{u}\|_2 \leq \|\mathbf{w}\|_2$, then $\|\mathbf{u}\|_{2, \Delta x} \leq \|\mathbf{w}\|_{2, \Delta x}$

New Norm: The norm $\|\mathbf{u}\| := \max_n \|\mathbf{u}^n\|_{2, \Delta x}$ where $\mathbf{u}^n = [u_1^n, u_2^n, \dots, u_{M-1}^n]^\top$
 Which means: Infinity norm in time, weighted 2-norm in space. 主要用于双变量 PDE

- For $\tau_m^n = \mathcal{O}((\Delta x)^\alpha) + \mathcal{O}((\Delta t)^b)$, $\Delta x = \frac{b-\Delta t}{M-\Delta t}$, $\Delta t = \frac{b}{N}$, assume $\Delta t = K(\Delta x)^\beta$, then:
 $\max_n \|\mathbf{x}^n\|_{2, \Delta x} = \mathcal{O}((\Delta x)^{\min\{\alpha, b\}})$

1.3 Linear Algebra

Definite: Positive Definite: $\lambda_i > 0, \forall i$ Negative Definite: $\lambda_i < 0, \forall i$

Symmetric: If A symmetric, then $A = Q \Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.

Normal: Real matrix A is normal if $A A^\top = A^\top A$ The following are properties of it:

1. A is orthogonally diagonalizable, i.e. $A = Q \Lambda Q^\top$ where Q orthogonal, Λ diagonal with eigenvalues.
2. Exists an eigenvectors of A : $\{\mathbf{v}_i\}$ s.t. $\mathbf{v}_i^\top \mathbf{v}_j = \delta_{ij}$ and Symmetric $\Rightarrow$ Normal

2 Global/Local Error, Convergence, Consistency, Stability

Exact & Approx. Sol: U_n : approx. n -th time-step sol; u_n : exact sol at n -th time-step.

Global Error $(e_n, \mathbf{e}(\Delta t))$: $e_n = U_n - u(t = n\Delta t)$ $\mathbf{e}(\Delta t) = [e_0, e_1, \dots, e_N]^\top$

Convergence: A numerical method is convergent if $\lim_{\Delta t \rightarrow 0} \|\mathbf{e}(\Delta t)\| = 0$

If $\lim_{\Delta t \rightarrow 0} \frac{\|\mathbf{e}(\Delta t)\|}{(\Delta t)^p} = C$ (i.e. $\|\mathbf{e}(\Delta t)\| = \mathcal{O}((\Delta t)^p)$) $\Rightarrow$ p -th order accurate

Local Truncation Error (τ_n) : 数值分析的离散迭代公式, 将 U_n 替换为 u_n 后的误差 (LHS - RHS).

- e.g. For $\frac{U_{n+1} - U_n}{\Delta t} = f(U_n)$, $\tau_{n+1} = \frac{U_{n+1} - u_{n+1}}{\Delta t} - f(u_n)$ $\tau_0 = 0$

- e.g. 我们可以把线性数值方法写成 $Au = b$, A matrix, $u = [U_1, \dots, U_{N-1}]^\top$.

Then: $\tau = A u_{\text{exact}} - b$ where $u_{\text{exact}} = [u_1, \dots, u_{N-1}]^\top$, $\tau = [\tau_1, \dots, \tau_{N-1}]^\top$

ps: $Ae = -\tau$ where $e = u - u_{\text{exact}}$ Define $\tau(\Delta t) = [\tau_1, \dots, \tau_{N-1}]^\top$

Consistency: If $\lim_{\Delta t \rightarrow 0} \|\tau(\Delta t)\| = 0$, the method is consistent.

Linear Equation: Goal: Finally change ODE/PDE into $Au = b$ where A is matrix, $u = [U_1, \dots, U_{N-1}]^\top$

Stability: A numerical method is stable if $\exists \Delta x_0$ s.t. For the Linear System $Au = b$:

1. $A(\Delta x)$ is invertible $\forall \Delta x < \Delta x_0$
2. $\|A(\Delta x)^{-1} \mathbf{w}\| \leq C \|\mathbf{w}\|$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$

Property: Consistent + Stable $\Rightarrow$ Convergent (same norm!) and $\|\mathbf{e}(\Delta t)\| = \mathcal{O}(\|\tau(\Delta t)\|)$

3 Intro & Method of Lines

Taylor Series: $f(t + \Delta t) = f(t) + f'(t)\Delta t + \frac{1}{2}f''(t)(\Delta t)^2 + \dots + \mathcal{O}((\Delta t)^N)$ ps: $u_{n+1} = u(n\Delta t + \Delta t)$

$f(t) = f(a) + f'(a)(t-a) + \frac{f''(a)}{2!}(t-a)^2 + \dots + \frac{f^{(N-1)}(a)}{(N-1)!}(t-a)^{N-1} + \mathcal{O}((t-a)^N)$

Euler Methods: Forward: $\frac{du}{dt}|_n \Delta t \approx \frac{U_{n+1} - U_n}{\Delta t}$; Backward: $\frac{du}{dt}|_n \Delta t \approx \frac{U_n - U_{n-1}}{\Delta t}$

Leapfrog: $\frac{du}{dt}|_n \Delta t \approx \frac{U_{n+1} - U_{n-1}}{2\Delta t}$ **Trapezoidal:** $\frac{1}{2} \left[\frac{du}{dt}|_n \Delta t + \frac{du}{dt}|_{n+1} \Delta t \right] \approx \frac{U_{n+1} - U_n}{\Delta t}$

Centred Difference: $\frac{d^2 u}{dt^2}|_n \Delta t \approx \frac{U_{n+1} - 2U_n + U_{n-1}}{(\Delta t)^2}$

Method of Lines: Using $U_m(t) \approx u(a + m\Delta x, t)$, $U_m^n \approx u(a + m\Delta x, n\Delta t)$ a 为空间起点, 不重要

1. For 1D: a. 先对空间 x 离散化, 使用 U_m 表达, 此时 $\frac{d}{dt} \mathbf{U} = \dots$ where $\mathbf{U} = [U_1(t), U_2(t), \dots, U_{M-1}(t)]^\top$
 b. 对时间 t 使用 ODE 离散化, 使用 U_m^n 表达, 得到 Linear Equation.

2. Example: Crank-Nicolson for Heat Eq. $u_t = \kappa u_{xx}$: 对 x 用 centred difference; 对 t 用 trapezoidal:

$$\frac{U_{m+1}^{n+1} - U_m^{n+1}}{\Delta t} = \frac{\kappa}{2} \left(\frac{U_{m+1}^{n+1} - 2U_m^{n+1} + U_{m-1}^{n+1}}{(\Delta x)^2} + \frac{U_{m+1}^n - 2U_m^n + U_{m-1}^n}{(\Delta x)^2} \right)$$

$$\Rightarrow -r U_{m-1}^{n+1} + (2 + 2r) U_m^{n+1} - r U_{m+1}^{n+1} = r U_{m-1}^n + (2 - 2r) U_m^n + r U_{m+1}^n, \quad r = \frac{\kappa \Delta t}{(\Delta x)^2}$$

Solve it: By iteration: e.g. $A \mathbf{U}^{n+1} = B \mathbf{U}^n$, where $\mathbf{U}^n = [U_1^n, U_2^n, \dots, U_{M-1}^n]^\top$.

Truncation for MoL: 对于 τ_m^n , 离散 x 后 U_m 替换为 u_m ; 对于 τ_m^n , 再离散 t 后 U_m^n 替换为 u_m^n .

e.g. For Crank-Nicolson above: $\tau_m = \frac{\partial u_m}{\partial t} - \kappa \frac{u_{m+1} - 2u_m + u_{m-1}}{(\Delta x)^2}$

e.g. For Crank-Nicolson above: $\tau_m^n = \frac{u_{m+1}^n - u_m^n}{\Delta t} - \frac{\kappa}{2}(\dots)$

4 Lax-Richtmyer Stability, von Neumann Analysis

Abs. Stability for ODEs: For ODEs $\frac{du}{dt} = A u$, u vector. A 1-step method for t (i.e. $U^{n+1} = B U^n$)

It's absolutely stable if $\|B \mathbf{w}\|_2 \leq \|\mathbf{w}\|_2$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$

Property: If B normal, then absolutely stable $\Leftrightarrow |\lambda_i(B)| \leq 1, \forall i$

Property: Absolute stability $\Rightarrow$ Lax-Richtmyer stability in 2-norm and weighted 2-norm

Lax-Richtmyer Stability: For PDE which discrete form $\frac{1}{\Delta t} U^{n+1} = \frac{1}{\Delta t} B U^n + F^n$, B fixed, F can change.

It's Lax-Richtmyer stable if $\|B^n \mathbf{w}\| \leq C \|\mathbf{w}\|$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$, $n \leq \frac{b}{\Delta t} = N$

Property: If consistent with $\max_n \|\mathbf{x}^n\|_{2, \Delta x} = \mathcal{O}((\Delta x)^\beta)$ and L-R stable $\|B^n \mathbf{w}\|_{2, \Delta x} \leq C \|\mathbf{w}\|_{2, \Delta x}$,

then, it converges with $\max_n \|\mathbf{e}^n\|_{2, \Delta x} = \mathcal{O}((\Delta x)^\beta)$ Remark: Always Assume $\Delta t = K(\Delta x)^\alpha$

von Neumann Analysis: PDE: 1. periodic BCs in spatial x , length D . 2. 满足 L-R 中 PDE 框架, 3. $F^n = 0$

$\Rightarrow$ Let $v_k, m = \exp(2\pi i k \frac{m \Delta x}{D})$ for $m = 0, 1, \dots, M-1$. Then vector $\mathbf{v}_k$ 为 B 的正交特征向量.

* By $U_m^n = \exp(2\pi i k \frac{m \Delta x}{D})$, $U_m^{n+1} = \alpha_k U_m^n$, 可得特征值(放大因子) α_k 表达式 (i.e. $B \mathbf{v}_k = \alpha_k \mathbf{v}_k$).

$\Rightarrow$ If all α_k for $k = 0, 1, \dots, M-1$ can be found $\Rightarrow B$ is normal and then:

If $|\alpha_k| \leq 1, \forall k \Rightarrow B$ is absolutely stable and L-R stable in weighted 2-norm

Remark: 假设 $\Delta t = K(\Delta x)^\alpha$ 会得 K, α 的限制条件, $\|$ For $F^n \neq 0$, 讨论 $F^n = 0$ 足够.

5 Boundary Condition

2nd ODE: For $u_x x = \beta(x)$ with $x \in (a, b)$, there are three types of BCs:

1. **Two Dirichlet:** $u(x=a) = c$ $u(x=b) = d$
2. **One Dirichlet, One Neumann:** $u(x=a) = c$ $u_x(x=b) = d$ or $u_x(x=a) = c$ $u(x=b) = d$
3. **One Neumann, One Constraint:** $u_x(x=a) = c$ or $\int_a^b u(x) dx = d$

Heat PDE: For $u_t = \kappa u_{xx}$ with $x \in (a, b)$, there are three types of BCs: * Basic IC: $u(x, t_0) = P(x)$

1. **Two Dirichlet:** $u(x=a, t) = c$ $u(x=b, t) = d$
2. **One Dirichlet, One Neumann:** $u(x=a, t) = c$ $u_x(x=b, t) = d$ or $u_x(x=a, t) = c$ $u(x=b, t) = d$
3. **Two Neumann:** $u_x(x=a, t) = c$ $u_x(x=b, t) = d$

Remark: For $u_x(x, a, t) = c \Rightarrow$ describe $\frac{u_m^n - u_0^n}{\Delta x} = c$ Boundary condition itself can effect the accuracy

For $u_t(x, t_0) = P(x) \Rightarrow$ describe $\frac{u_m^n - u_0^n}{\Delta t} = c$ (a + m \Delta x)

6 One Time Solve Two Variables PDEs

Elliptic Equation: $L u = F(x_1, \dots, x_d)$ is an elliptic PDE if L is an elliptic operator.

Theorem: $L = \sum_{r=1}^d \sum s = 1^d A_{r,s} \frac{\partial^2}{\partial x_r \partial x_s} + \sum_{n=1}^d B_n \frac{\partial}{\partial x_n} + C$ is elliptic operator if:

A is symmetric and A is positive/negative definite ($\lambda_i > 0 | \lambda_i < 0$).

Poisson Equation: $\Delta u = F(x_1, \dots, x_d)$ is a Poisson equation and it's elliptic. with BCs:

Dirichlet: $u = u_D$ on $\partial \Omega$; Neumann: $\Delta u \cdot \hat{\mathbf{n}} = g$ on $\partial \Omega$ where $\int_{\Omega} F = \int_{\partial \Omega} g$ and $\int_{\Omega} u = \int_{\partial \Omega} u_D$.

e.g. For 2D: $u_{xx} + u_{yy} = F(x, y)$ with BCs: $u(x, y) = u_D$ on $\partial \Omega$

$\rightarrow$ **5-point Laplacian:** $\frac{U_{m-1,p} + U_{m+1,p} + U_{m,p-1} + U_{m,p+1} - 4U_{m,p}}{(\Delta x)^2} = F(a + m \Delta x, a + p \Delta x)$

$\rightarrow$ Using $\mathbf{u} = [U_{1,1}, U_{2,1}, \dots, U_{M-1,1}, U_{1,2}, \dots, U_{M-1,M-1}]^\top$ and

$\mathbf{b} = (\mathbf{u} - \mathbf{p}) - 1(M-1) = F(a + m \Delta x, a + p \Delta x)$ then $A \mathbf{u} = \mathbf{b}$ where A is a symmetric matrix.

$\rightarrow$ Eigenvalue/vectors of A : $\mathbf{v}_{k,l} = [v_{k,l,1,1}, v_{k,l,1,2}, \dots, v_{k,l,1,2}, \dots, v_{k,l,M-1,M-1}]^\top$
 $v_{k,l,m,p} = \sin(\pi k m \Delta x) \sin(\pi l p \Delta x)$, $\lambda_{k,l} = \frac{2}{(\Delta x)^2} [\cos(\pi k \Delta x) + \cos(\pi l \Delta x) - 2] < 0$, $\lambda_{\min} = \lambda_{1,1}$

Remark: 这一节都是用同一个 Δx 离散 x 和 y 的, 和前面 t, x 分别不同.

Consistency: If $\lim_{\Delta x \rightarrow 0} \frac{\|\tau(\Delta x)\|}{\Delta x} = 0$, the method is consistent.

Convergence: If $\|\mathbf{e}(\Delta x)\| = \mathcal{O}((\Delta x)^p)$, the method is p -th order accurate.

Stability: A numerical method is stable if for the linear system $A \mathbf{u} = \mathbf{b}$:

1. $A(\Delta x)$ is invertible $\forall \Delta x < \Delta x_0$
2. $\|A(\Delta x)^{-1} \mathbf{w}\| \leq C \|\mathbf{w}\|$ for all $\mathbf{w} \in \mathbb{R}^{N-1}$

Property: If A normal, then $\|A^{-1} \mathbf{w}\|_2 \leq \frac{1}{|\lambda_1|} \|\mathbf{w}\|_2$ where $0 < |\lambda_1| \leq \dots \leq |\lambda_{N-1}|$ are eigenvalues of A .

7 Spectral Methods

Inner Product Space for Function: $\langle \phi_i, \phi_j \rangle = \int_a^b \phi_i \phi_j w dx$ ps: $w(x)$ is weight function.

For Fourier Series: $w(x) = 1, \alpha = -\pi, b = \pi$: $\sin(n x), \cos(m x)$ are orthogonal basis.

Fourier Series: $f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} a_n \cos(n x) + b_n \sin(n x) = \sum_{n=-\infty}^{\infty} A_k e^{i k x}$ where

$\rightarrow a_0 = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx$ $a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos(n x) dx$ $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin(n x) dx$.

$\rightarrow k > 0: A_k = \frac{1}{2}(\alpha_k - i b_k); k < 0: A_k = \frac{1}{2}(\alpha_{-k} + i b_{-k}); k = 0: A_0 = \frac{a_0}{2}$. For real func: $A_{-k} = \overline{A_k}$

Property: $A_k = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-i k x} dx$ and $f'(x) = \sum_{k=-\infty}^{\infty} (i k) A_k e^{i k x}$

More on Fourier Series: Assume f is periodic with period 2π , then:

1. The Fourier Transform: $\mathcal{F}\{f(x)\} = \hat{f}(k) = \frac{1}{2\pi} \int_{-\pi}^{\pi} f(x) e^{-i k x} dx$

2. The inverse Fourier Transform: $\mathcal{F}^{-1}\{\hat{f}(k)\} = f(x) = \sum_{k=-\infty}^{\infty} \hat{f}(k) e^{i k x}$

Change $L \rightarrow 2\pi$ period: If $f(x)$ has period $L \Rightarrow g(x) = f(\frac{x}{2\pi} L)$ has 2π period.

Discrete Fourier Transform (DFT): If $x_n = n \Delta x$ equally-spaced with $n = 0, 1, \dots, N-1; \Delta x = \frac{2\pi}{N}$:

1. **Fourier Transform:** $\mathcal{F}\{f\} = \hat{f}_k = \frac{2\pi}{N} \sum_{n=0}^{N-1} f(x_n) e^{-i k x_n}$ $k = -\frac{N}{2} + 1, \dots, \frac{N}{2}$

2. **Inverse FT:** $\mathcal{F}^{-1}\{\hat{f}\} = f(x) = \frac{1}{2\pi} \sum_{k=-N/2+1}^{N/2} \hat{f}_k e^{i k x_n} = \frac{1}{2\pi} \sum_{k=-N/2}^{N/2} \hat{f}_k e^{i k x_n}$

where $\hat{f}_k = \hat{f}_k$ for $k = -\frac{N}{2} + 1, \dots, \frac{N}{2} - 1; \hat{f}_{-N/2} = \hat{f}_{N/2} = \frac{1}{2} \hat{f}_{N/2}$.

Property: $\hat{f}'_k = (i k) \hat{f}_k \Leftrightarrow \mathcal{F}\{f'\} = (i k) \mathcal{F}\{f\}$

Remark: Quickly compute DFT: Fast Fourier Transform (FFT)

Using Fourier to solve PDE: For PDE with periodic BCs

1. Take Fourier Transform in PDE (in space) e.g. $u_t = u_{xx}$ $\rightarrow \frac{d}{dt} \hat{u}_k = -k^2 \hat{u}_k$
2. Take Fourier Transform in IC (in space) e.g. $u(x, 0) = f(x) \rightarrow \hat{u}_k(t=0) = \hat{f}_k$
3. Using Transformed PDE and IC $\rightarrow$ a. Directly Solve ODE. b. numerical method for ODE. $\rightarrow \hat{u}_k(t)$
4. Inverse Fourier Transform to get $u(x_n, t)$ 对 x_n 点 t 的估计或精确解.

Ideal of Interpolants: 给定 $(x_n, f(x_n))$, 构造插值函数 $p(x)$ s.t. $p(x_n) = f(x_n) \Rightarrow$ 用 $p'(x_n)$ 估计 $f'(x_n)$.

Spectral Collocation: Given $(x_n, f(x_n))$, f has period 2π , construct Fourier Interpolant $p(x)$ s.t. $p(x_n) = f(x_n)$

1. DFT: $\hat{f}_k = \frac{2\pi}{N} \sum_{n=0}^{N-1} f(x_n) e^{-i k x_n} \Rightarrow p(x) = \frac{1}{2\pi} \sum_{k=-N/2}^{N/2} \hat{f}_k e^{i k x}$

2. (Method I) (即 FFT, 无法得到微分算子的矩阵) Direct differentiate $p(x)$, and $f'(x_n) \approx p'(x_n)$.

3. (Method II) Assume $p(x) = \sum_{m=0}^{N-1} f(x_m) P_5(x - x_m)$ where $P_5(x_j - x_m) = \delta_{j-m}$
 $\Rightarrow P_5(x - x_m) = \frac{1}{2\pi} \sum_{k=-N/2}^{N/2} \hat{P}_{5,k} e^{i k x} = \frac{1}{2\pi} \sum_{k=-N/2}^{N/2} e^{i k(x-x_m)} \text{ and } P_5(x) = \frac{\sin(\pi x / \Delta x)}{(2\pi / \Delta x) \tan(x/2)}$

$\Rightarrow f'(x_n) \approx p'(x_n) = \sum_{m=0}^{N-1} f(x_m) P'_5(x_n - x_m)$ and $P'_5(x_n - x_m) = \begin{cases} 0 & j \equiv 0 \pmod{N} \\ \frac{1}{2}(-1)^j \cot(\frac{i h}{2}) & j \not\equiv 0 \pmod{N} \end{cases}$

10 The Steepest Descent Method

Theorem: Let $\phi(\mathbf{x}) = \frac{1}{2} \mathbf{x}^T \mathbf{A} \mathbf{x} - \mathbf{b}^T \mathbf{x}$, then it has unique minimum if $\mathbf{A}$ is *positive definite*.
If $\mathbf{A}$ is symmetric positive definite $\Rightarrow \mathbf{x}:\mathbf{Ax} = \mathbf{b} \Leftrightarrow \mathbf{x}:\phi(\mathbf{x}) = \min_{\tilde{\mathbf{x}}} \phi(\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}} [\frac{1}{2} \tilde{\mathbf{x}}^T \mathbf{A} \tilde{\mathbf{x}} - \mathbf{b}^T \tilde{\mathbf{x}}]$
Following need $\mathbf{A}$ to be symmetric positive definite.

Steepest Descent Method: To solve $\min \phi(x)$ or $\mathbf{Ax} = \mathbf{b}$, we can use the following iteration:
 $\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} - \alpha_{k-1} \nabla \phi(\mathbf{x}^{(k-1)})$ and α_{k-1} s.t. $\min_{\alpha} \phi(\mathbf{x}^{(k-1)} - \alpha \nabla \phi(\mathbf{x}^{(k-1)}))$.

- Property 1:** $\nabla \phi(\mathbf{x})$ is the direction of steepest ascent (most rapid increase) of ϕ at $\mathbf{x}$.
- Property 2:** $\alpha_{k-1} = \frac{(\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)}}{(\mathbf{r}^{(k-1)})^T \mathbf{A} \mathbf{r}^{(k-1)}}$ and $\nabla \phi(\mathbf{x}^{(k)}) = \mathbf{Ax}^{(k)} - \mathbf{b} =: -\mathbf{r}^{(k)}$ is k -th residual.
- Property 3:** $\mathbf{r}^{(k)} = \mathbf{b} - \mathbf{Ax}^k = \dots = \mathbf{r}^{(k-1)} - \alpha_{k-1} \mathbf{Ar}^{(k-1)}$

Remark: Convergence behaviour is *highly problem-dependent*. (如果 ϕ 的等高线是圆形则快, 椭圆形则慢)
The Conjugate Gradient Method: To solve $\min \phi(x)$ or $\mathbf{Ax} = \mathbf{b}$, we can use the following iteration:
 $\mathbf{x}^{(k)} = \mathbf{x}^{(k-1)} + \alpha_{k-1} \mathbf{p}^{(k-1)}$ and α_{k-1} s.t. $\min_{\alpha} \phi(\mathbf{x}^{(k-1)} + \alpha \mathbf{p}^{(k-1)})$
and $\mathbf{p}^{(k)}$ is $\mathbf{A}$ -conjugate to previous **search direction**: $\mathbf{p}^{(k-1)} : (\mathbf{p}^{(k)})^T \mathbf{A} \mathbf{p}^{(k-1)} = 0$

- Property 1:** $\alpha_{k-1} = \frac{(\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)}}{(\mathbf{p}^{(k-1)})^T \mathbf{A} \mathbf{p}^{(k-1)}}$ and $-\mathbf{r}^{(k)} = \mathbf{Ax}^{(k)} - \mathbf{b}$ is k -th residual.
- Property 2:** $\mathbf{p}^{(k)} = \mathbf{r}^{(k)} + \beta_{k-1} \mathbf{p}^{(k-1)}$ where $\beta_{k-1} = \frac{(\mathbf{r}^{(k)})^T \mathbf{r}^{(k)}}{(\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)}}$
- Property 3:** $(\mathbf{p}^{(k)})^T \mathbf{A} \mathbf{p}^{(j)} = 0$ and $(\mathbf{r}^{(k)})^T \mathbf{r}^{(j)} = 0$ for $j = 0, 1, \dots, k-1$
- Property 4:** $\text{span}\{\mathbf{p}^{(0)}, \dots, \mathbf{p}^{(k-1)}\} = \text{span}\{\mathbf{A}^0 \mathbf{r}^{(0)}, \dots, \mathbf{A}^{k-1} \mathbf{r}^{(0)}\} = \text{span}\{\mathbf{A}^0 \mathbf{e}^{(0)}, \dots, \mathbf{A}^k \mathbf{e}^{(0)}\}$
where $\mathbf{e}^{(k)} = \mathbf{x}^{(k)} - \mathbf{x}$ is the error at k -th iteration. (Quicker than JAC/GAU)
- Exact Property:** For $\mathbf{A} \in \mathbb{R}$ has n distinct eigenvalues $\Rightarrow$ solve in at **most** n iterations. (By exact arithmetic)

A-Norm: The $\mathbf{A}$ -norm of $\mathbf{x}$ is defined as $\|\mathbf{x}\|_{\mathbf{A}} = \sqrt{\mathbf{x}^T \mathbf{A} \mathbf{x}}$.
Condition Number: The condition number of $\mathbf{A}$ is $\kappa(\mathbf{A}) = \|\mathbf{A}\| \|\mathbf{A}^{-1}\| \geq 1$ and $\kappa_2(\mathbf{A}) = \frac{\lambda_{\max}(\mathbf{A})}{\lambda_{\min}(\mathbf{A})}$
Error of Conjugate Gradient: To solve $\mathbf{Ax} = \mathbf{b}$, error is: $\frac{\|\mathbf{e}^{(k)}\|_{\mathbf{A}}}{\|\mathbf{e}^{(0)}\|_{\mathbf{A}}} \leq 2 \left(\frac{\sqrt{\kappa(\mathbf{A})}-1}{\sqrt{\kappa(\mathbf{A})}+1} \right)^k$

11 The advection equation

Hyperbolic system: For $\mathbf{u}_t + \mathbf{Au}_x = 0$, if $\mathbf{A}$ is diagonalizable with real eigenvalues $\Rightarrow$ hyperbolic system.
Diagonalization: If $P^{-1} \mathbf{A} P = \Lambda$, then Let $\mathbf{v} = P^{-1} \mathbf{u}$, we have $\mathbf{v}_t + \Lambda \mathbf{v}_x = 0$
Advection Equation: The advection equation is $u_t + v u_x = 0$ where v is the advection speed.
Translate to Wave Equation: 两边对 t 求导后代入原式可得 $u_{tt} = v^2 u_{xx}$
-If IC is $u(x, 0) = P(x)$: solution is $u(x, t) = P(x - vt)$
-Moreover, if BC is Periodic: sol $u(x, t) = \tilde{P}(x - vt)$ where $\tilde{P}(x + k(b-a)) = P(x)$ for $x \in [a, b], k \in \mathbb{Z}$.
-If BC is $u(a, t) = P(t)$: solution is $u(x, t) = P(t - \frac{x-a}{v})$
Remark: Periodic BCs $\Rightarrow U_p = U_{M+p}$ for $p = -1, 0$

Courant Number: The Courant number is defined as $C = \frac{v \Delta t}{\Delta x}$.
Characteristics Based Method: For the advection equation $u_t + v u_x = 0$, and $|C| = |\frac{v \Delta t}{\Delta x}| \leq 1$:
 $\Rightarrow u(x_m, t_{n+1}) = u(x_m - v \Delta t, t_n) \approx U_m^n + \frac{|v|+v}{2} \frac{\Delta t}{\Delta x} (U_m^{n-1} - U_m^n) + \frac{|v|-v}{2} \frac{\Delta t}{\Delta x} (U_{m+1}^n - U_m^n) =: U_m^{n+1}$
Equivalently, $\frac{U_m^{n+1} - U_m^n}{\Delta t} + \frac{|v|+v}{2} \frac{U_m^n - U_{m-1}^n}{\Delta x} - \frac{|v|-v}{2} \frac{U_{m+1}^n - U_m^n}{\Delta x} = 0$
Remark: Above is for linear interpolation. (e.g. For $v > 0, u(x_m, t_{n+1}) \approx U_{n+1}^n = (1-C)U_m^n + CU_{m-1}^n$)
Upwinding: For the advection equation $u_t + v u_x = 0$:
For $v > 0$, use $U_m^n, U_{m-1}^n, U_{m-2}^n, \dots$ to estimate $u(x_m - v \Delta t, t_n) = u(x_m, t_{n+1})$, denote by U_m^{n+1} .
For $v < 0$, use $U_m^n, U_{m+1}^n, U_{m+2}^n, \dots$ to estimate $u(x_m - v \Delta t, t_n) = u(x_m, t_{n+1})$, denote by U_m^{n+1} .
 $\Rightarrow$ Thus $v > 0 \Rightarrow \frac{dU_m}{dt} + \frac{v}{\Delta x} \sum_{q=0}^Q \gamma_q U_{m-q} = 0$ (i.e. $\frac{du}{dx} |_{x_m} \approx \frac{1}{\Delta x} \sum_{q=0}^Q \gamma_q U_{m-q}$)
 $\Rightarrow$ Thus $v < 0 \Rightarrow \frac{dU_m}{dt} + \frac{v}{\Delta x} \sum_{q=0}^Q \gamma_q U_{m+q} = 0$ (i.e. $\frac{du}{dx} |_{x_m} \approx \frac{1}{\Delta x} \sum_{q=0}^Q \gamma_q U_{m+q}$)
ps: For $u_t + v u_x = 0 \Rightarrow$ In characteristics form: $\frac{du}{dt} = 0$ on $\frac{dx}{dt} = v \Rightarrow u = \mathbf{A}$ on $x = vt + B$

12 Appendix

12.1 Other
Divergence Theorem: $\int_{\Omega} \nabla \cdot \mathbf{F} dV = \int_{\partial \Omega} \mathbf{F} \cdot \hat{\mathbf{n}} dS$ $\hat{\mathbf{n}}$ outward normal. $\nabla^2 = \nabla \cdot \nabla = \Delta$
How to improve Approximation Accuracy?: a. Using more neighboring points. (more terms from Taylor expansion) b. Make $\Delta t, \Delta x$ smaller. c. For Periodic BCs, use Fourier Transform.
12.2 ODE
 $y' + P y = Q; \mu y = \int Q(x) \mu dx + C, \mu(x) = e^{\int P(x) dx} \quad y' = P y; y = C e^{\int P(x) dx}$
 $y'' + k^2 y = 0; y = A \cos(kx) + B \sin(kx) \quad y'' - k^2 y = 0; y = A e^{kx} + B e^{-kx}, k > 0$
 $\alpha y'' + b y' + c y = 0$: Characteristic equation: $\alpha r^2 + b r + c = 0 \quad r_{1,2} = \mu \pm i \nu$
a. $\Delta > 0: y = A e^{r_1 x} + B e^{r_2 x}$; b. $\Delta = 0: y = (A + B x) e^{\mu x}$; c. $\Delta < 0: y = e^{\mu x} (A \cos(\nu x) + B \sin(\nu x))$
12.3 Integral
Gaussian Integral: $\int_{-\infty}^{\infty} e^{-ax^2} dx = \sqrt{\frac{\pi}{a}} \quad \int_0^{\infty} e^{-x t^n} t^m = \frac{1}{n} \Gamma(\frac{m+1}{n}) x^{-\frac{m+1}{n}}$
Integral: $\int \tan x = -\ln |\cos x| = \ln |\sec x| \quad \int \cot x = \ln |\sin x| = -\ln |\csc x|$
Techniques: a. e.g. $\int e^{at} \cos(bt) dt = \text{Re}(\int e^{(a+ib)t} dt)$
Basis: For $(\cdot, \cdot)$ with $(f, g) = \int_a^b f(x) g(x) dx$ ($\rho = 1$) we have: $(\sin(n\pi(x-a)/L), \cos(m\pi(x-b)/L)) = 0$
For $(\sin(n\pi(x-a)/L), \sin(m\pi(x-b)/L)) = \frac{\pi}{2} \delta_{nm}, (\cos(n\pi(x-a)/L), \cos(m\pi(x-b)/L)) = \frac{\pi}{2} \delta_{nm}$.

12.4 Expansion
Binomial Expansion: $(1+x)^\alpha \sim 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!} x^2 + \dots$ as $x \rightarrow 0$
For $x \rightarrow a$: Taylor at $x = a: f(x) \sim f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \dots$ as $x \rightarrow a$
Taylor $[x \rightarrow 0]$: $f(x+h) \sim f(x) + f'(x)h + \frac{f''(x)}{2!} h^2 + \dots$ as $h \rightarrow 0$
 $\sin(x) \sim x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots$ as $x \rightarrow 0 \quad e^x \sim 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots$ as $x \rightarrow 0$
 $\cos(x) \sim 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \dots$ as $x \rightarrow 0 \quad \ln(1+x) \sim x - \frac{x^2}{2} + \frac{x^3}{3} - \dots$ as $x \rightarrow 0$
 $\sinh(x) \sim x + \frac{x^3}{3!} + \frac{x^5}{5!} + \dots$ as $x \rightarrow 0 \quad \cosh(x) \sim 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \dots$ as $x \rightarrow 0$
 $\tan(x) \sim x + \frac{x^3}{3} + \frac{2x^5}{15} + \dots$ as $x \rightarrow 0 \quad \cot(x) \sim \frac{1}{x} - \frac{\pi}{3} - \frac{x^3}{45} - \dots$ as $x \rightarrow 0$
For $x \rightarrow \infty$: $\arctan(x) \sim \frac{\pi}{2} - \frac{1}{x} + \frac{1}{3x^3} - \dots$ as $x \rightarrow \infty$
 $\ln(a+x) = \ln x + \ln(1 + \frac{a}{x}) \sim \ln x + \frac{a}{x} - \frac{a^2}{2x^2} + \dots$ as $x \rightarrow \infty$
 $\frac{1}{a+x} = \frac{1}{x} \cdot \frac{1}{1+\frac{a}{x}} \sim \frac{1}{x} - \frac{a}{x^2} + \frac{a^2}{x^3} - \dots$ as $x \rightarrow \infty$
sin|cos: $a \sin(x) + b \cos(x) = \sqrt{a^2 + b^2} \sin(x + \varphi) \quad a \cos(x) + b \sin(x) = \sqrt{a^2 + b^2} \cos(x - \varphi), \varphi \text{ 过 } (a, b).$
 $-\sin(x) + \sin(y) = 2 \sin(\frac{x-y}{2}) \cos(\frac{x+y}{2}) \mid \sin(x) \cos(y) = \frac{1}{2} [\sin(x+y) + \sin(x-y)] \quad S+S=CS, S-S=CS, C+C=CC, C-C=-SS$
Inequalities: $\ln(x+1) < x, e^x > x+1, (1+x)^a < e^{ax}$
12.5 Other
Nabla: $\nabla^2 u = u_{xx} + u_{yy} + u_{zz} + \dots \quad \nabla \cdot \mathbf{F} = \frac{\partial F_x}{\partial x} + \frac{\partial F_y}{\partial y} + \frac{\partial F_z}{\partial z}$
Binomial Expansion: $(a+b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k \quad \binom{n}{k} = \frac{n!}{k!(n-k)!} \quad (1+x)^n = \sum_{k=0}^n \binom{n}{k} x^k$
Sin/Cos: $\sin[\frac{2n+1}{2}\pi] = (-1)^n \quad \cos(n\pi) = (-1)^n$

13 Some Proofs

#10 Theorem Proof: $\phi(\mathbf{x}) = \min_{\tilde{\mathbf{x}}} \phi(\tilde{\mathbf{x}}) = \min_{\tilde{\mathbf{x}}} [\frac{1}{2} \tilde{\mathbf{x}}^T \mathbf{A} \tilde{\mathbf{x}} - \mathbf{b}^T \tilde{\mathbf{x}}]$. A symmetric positive definite, $\mathbf{x}$ solves $\mathbf{Ax} = \mathbf{b}$.
 $(\Rightarrow)$ Let $\mathbf{x}$ solve $\mathbf{Ax} = \mathbf{b}$, then $\phi(\mathbf{x}) = \frac{1}{2} \underbrace{\left(\mathbf{A} \frac{1}{2} \mathbf{x} - \mathbf{A}^{-\frac{1}{2}} \mathbf{b} \right)^T \left(\mathbf{A} \frac{1}{2} \mathbf{x} - \mathbf{A}^{-\frac{1}{2}} \mathbf{b} \right)}_{=|\mathbf{A}^{1/2} \mathbf{x} - \mathbf{A}^{-1/2} \mathbf{b}|_2^2} - \underbrace{\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}}_{\text{constant}}$
As $\mathbf{Ax} = \mathbf{b}$, then $\mathbf{A} \frac{1}{2} \mathbf{x} - \mathbf{A}^{-\frac{1}{2}} \mathbf{b} = 0$, thus $\phi(\mathbf{x}) = -\frac{1}{2} \mathbf{b}^T \mathbf{A}^{-1} \mathbf{b}$ achieves the minimum.
 $(\Leftarrow)$ Let $\phi(\mathbf{x}) = \min_{\tilde{\mathbf{x}}} \phi(\tilde{\mathbf{x}})$, then $\nabla \phi(\mathbf{x}) = \mathbf{Ax} - \mathbf{b} = 0$, thus $\mathbf{Ax} = \mathbf{b}$.
#10 Why choose α_{k-1} s.t. $\min_{\alpha} \phi(\mathbf{x}^{(k-1)} + \alpha \mathbf{p}^{(k-1)})$: Steepest descent method
Since $\phi(\mathbf{x}^{(k-1)} + \alpha \mathbf{r}^{(k-1)}) = [\frac{1}{2} (\mathbf{x}^{(k-1)})^T \mathbf{A} \mathbf{x}^{(k-1)} - \mathbf{b}^T \mathbf{x}^{(k-1)}] + \alpha [\mathbf{x}^{(k-1)}]^T \mathbf{A} \mathbf{r}^{(k-1)} - \mathbf{b}^T \mathbf{r}^{(k-1)}] + \frac{\alpha^2}{2} (\mathbf{r}^{(k-1)})^T \mathbf{A} \mathbf{r}^{(k-1)}$
So that $\frac{d\phi}{d\alpha} = (\mathbf{x}^{(k-1)})^T \mathbf{A} \mathbf{r}^{(k-1)} - \mathbf{b}^T \mathbf{r}^{(k-1)} + \alpha (\mathbf{r}^{(k-1)})^T \mathbf{A} \mathbf{r}^{(k-1)}$
 $= (\mathbf{Ax}^{(k-1)} - \mathbf{b})^T \mathbf{r}^{(k-1)} + \alpha (\mathbf{r}^{(k-1)})^T \mathbf{A} \mathbf{r}^{(k-1)} = -(\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)} + \alpha (\mathbf{r}^{(k-1)})^T \mathbf{A} \mathbf{r}^{(k-1)}$
Set $\frac{d\phi}{d\alpha} = 0$, we have $\alpha_{k-1} = \frac{(\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)}}{(\mathbf{r}^{(k-1)})^T \mathbf{A} \mathbf{r}^{(k-1)}}$
#10 Why choose α_{k-1} s.t. $\min_{\alpha} \phi(\mathbf{x}^{(k-1)} + \alpha \mathbf{p}^{(k-1)})$: Conjugate gradient method
Since $\phi(\mathbf{x}^{(k-1)} + \alpha \mathbf{p}^{(k-1)}) = [\frac{1}{2} (\mathbf{x}^{(k-1)})^T \mathbf{A} \mathbf{x}^{(k-1)} - \mathbf{b}^T \mathbf{x}^{(k-1)}] + \alpha [\mathbf{x}^{(k-1)}]^T \mathbf{A} \mathbf{p}^{(k-1)} - \mathbf{b}^T \mathbf{p}^{(k-1)}] + \frac{\alpha^2}{2} (\mathbf{p}^{(k-1)})^T \mathbf{A} \mathbf{p}^{(k-1)}$
So that $\frac{d\phi}{d\alpha} = (\mathbf{x}^{(k-1)})^T \mathbf{A} \mathbf{p}^{(k-1)} - \mathbf{b}^T \mathbf{p}^{(k-1)} + \alpha (\mathbf{p}^{(k-1)})^T \mathbf{A} \mathbf{p}^{(k-1)}$
 $= (\mathbf{Ax}^{(k-1)} - \mathbf{b})^T \mathbf{p}^{(k-1)} + \alpha (\mathbf{p}^{(k-1)})^T \mathbf{A} \mathbf{p}^{(k-1)} = -(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-1)} + \alpha (\mathbf{p}^{(k-1)})^T \mathbf{A} \mathbf{p}^{(k-1)}$
Set $\frac{d\phi}{d\alpha} = 0$, we have $\alpha_{k-1} = \frac{(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-1)}}{(\mathbf{p}^{(k-1)})^T \mathbf{A} \mathbf{p}^{(k-1)}}$ As $(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-1)} = (\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)}$ so we have $\alpha_{k-1} = \dots$
Show that: $(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-1)} = (\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)}$.
Since $\mathbf{p}^{(k)} = \mathbf{r}^{(k)} + \beta_{k-1} \mathbf{p}^{(k-1)}$, we have $(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-1)} = (\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)} + \beta_{k-2} (\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-2)}$.
As $(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-2)} = 0$, we have $(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-1)} = (\mathbf{r}^{(k-1)})^T \mathbf{r}^{(k-1)}$.
Show that: $(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-2)} = 0$.
Since $\phi(\alpha_{k-2}) = \frac{1}{2} (\mathbf{x}^{(k-2)} + \alpha_{k-2} \mathbf{p}^{(k-2)})^T \mathbf{A} (\mathbf{x}^{(k-2)} + \alpha_{k-2} \mathbf{p}^{(k-2)}) - \mathbf{b}^T (\mathbf{x}^{(k-2)} + \alpha_{k-2} \mathbf{p}^{(k-2)})$.
By using $\mathbf{A} = \mathbf{A}^T$, we have $\phi'(\alpha_{k-2}) = (\mathbf{x}^{(k-2)})^T \mathbf{A} \mathbf{p}^{(k-2)} - \mathbf{b}^T \mathbf{p}^{(k-2)} + \alpha_{k-2} (\mathbf{p}^{(k-2)})^T \mathbf{A} \mathbf{p}^{(k-2)}$
 $= (\mathbf{Ax}^{(k-2)} - \mathbf{b})^T \mathbf{p}^{(k-2)} + \alpha_{k-2} (\mathbf{p}^{(k-2)})^T \mathbf{A} \mathbf{p}^{(k-2)}$
Since $\phi'(\alpha_{k-2}) = 0$, we have $(\mathbf{Ax}^{(k-2)} - \mathbf{b})^T \mathbf{p}^{(k-2)} + \alpha_{k-2} (\mathbf{p}^{(k-2)})^T \mathbf{A} \mathbf{p}^{(k-2)} = 0$.
So that $(\mathbf{Ax}^{(k-2)} + \alpha_{k-2} \mathbf{p}^{(k-2)}) - \mathbf{b}^T \mathbf{p}^{(k-2)} = 0$.
As $\mathbf{x}^{(k-1)} = \mathbf{x}^{(k-2)} + \alpha_{k-2} \mathbf{p}^{(k-2)}$ and $-\mathbf{r}^{(k-1)} = \mathbf{Ax}^{(k-1)} - \mathbf{b}$, we have $(\mathbf{r}^{(k-1)})^T \mathbf{p}^{(k-2)} = 0$.
#1.2 Norm: For $d = 1, \Delta x = \frac{D}{N}, u_i = \mathcal{O}((\Delta x)^P)$, then $\|\mathbf{u}\|_2, \Delta x = \mathcal{O}((\Delta x)^P)$.
Proof: $\|\mathbf{u}\|_{2, \Delta x}^2 = \Delta x \sum_{i=1}^{N-1} u_i^2 \leq \Delta x (N-1) \max_{1 \leq i \leq N-1} u_i^2 \leq D \max_{1 \leq i \leq N-1} u_i^2$, where $\Delta x = \frac{D}{N}$.
Thus $\|\mathbf{u}\|_2, \Delta x = \mathcal{O}((\Delta x)^P)$.
Proof: If $\mathbf{u} = [U_{1,1}, U_{2,1}, \dots, U_{1,2}, U_{2,2}, \dots, U_{M-1,M-1}]^T, d = 2 (xy), x = m \Delta x, y = p \Delta x$,
 $\|\mathbf{u}\|_{2, \Delta x}^2 = (\Delta x)^2 \sum_{m=1}^{N-1} \sum_{p=1}^{M-1} U_{m,p}^2, p \leq (\Delta x)^2 (M-1)^2 \max_{1 \leq m, p \leq M-1} U_{m,p}^2$
 $\leq D^2 \max_{1 \leq i, j \leq M-1} U_{i,j}^2$, where $\Delta x = \frac{D}{M}$.
Thus $\|\mathbf{u}\|_2, \Delta x = \mathcal{O}((\Delta x)^P)$.

固定答题模板: It's given that the discretisation proposed above is stable in the 2-norm. Explain, step-by-step, how you would argue that this scheme is converges at $[n]$ -th order, in a norm which you should specify. [You are not repired to carry out the calculations,...]
Answer: The assumption given in part ([?]) tells us that the scheme is also stable in the **weighted 2-norm**. One would then analyse the local truncation error of the scheme finite difference scheme by using Taylor expansion. After algebraic calculations, and use of the partial differential equation itself. Notice that the local truncation error associated with the boundary conditions is zero, as this is applied exactly. Overall we could argue that the local truncation error is $\mathcal{O}((\Delta t)^P + (\Delta x)^Q)$, hence the scheme is consistent in the **weighted 2-norm**. Combining the stability and consistency results, we can conclude that the scheme is convergent in the **weighted 2-norm**.
or: **weighted 2-norm in x -dimension and ∞ -norm in t -dimension.**